Light

GCSE Computer Science — Programming Fundamentals

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a variable in programming?

- ○ **A.** A named location in memory that stores a value which can change
- ○ **B.** A fixed value that cannot be changed during program execution
- ○ **C.** A section of code that performs a specific task
- ○ **D.** A message displayed to the user on screen

Q2 (1 mark)

Why would a programmer use a constant instead of a variable?

- ○ **A.** To store values that need to change frequently
- ○ **B.** To make the program run faster
- ○ **C.** To reduce the amount of memory used
- ○ **D.** To store values that should remain fixed throughout the program

Q3 (1 mark)

Which data type would be most appropriate for storing a person's age?

- ○ **A.** Integer
- ○ **B.** String
- ○ **C.** Boolean
- ○ **D.** Real

Q4 (1 mark)

What values can a Boolean variable hold?

- ○ **A.** Any whole number
- ○ **B.** Any text string
- ○ **C.** Any number with decimal places
- ○ **D.** True or False only

Q5 (1 mark)

What is the purpose of comments in program code?

- ○ **A.** To explain what the code does for human readers
- ○ **B.** To make the program execute faster
- ○ **C.** To display messages to the user
- ○ **D.** To store temporary values during execution

Q6 (1 mark)

What type of error prevents a program from running at all?

- ○ **A.** Logic error
- ○ **B.** Runtime error
- ○ **C.** Semantic error
- ○ **D.** Syntax error

Q7 (1 mark)

A program runs without error messages but produces incorrect output. What type of error is this?

- ○ **A.** Syntax error
- ○ **B.** Logic error
- ○ **C.** Runtime error

- **D.** Compilation error

Q8 (1 mark)

What is the result of the expression `17 // 5` in Python?

- **A.** 3
- **B.** 3.4
- **C.** 2
- **D.** 4

Q9 (1 mark)

What does the modulus operator (%) return?

- **A.** The quotient of a division
- **B.** The percentage of one number relative to another
- **C.** The result of multiplying two numbers
- **D.** The remainder after integer division

Q10 (1 mark)

Which operator tests whether two values are equal?

- **A.** ==

- B. =
- C. !=
- D. <>

Q11 (1 mark)

What does the `input()` function do in Python?

- A. Displays a message on the screen
- B. Reads data from a file
- C. Generates a random number
- D. Waits for the user to enter data from the keyboard

Q12 (1 mark)

What programming construct allows a program to make decisions based on conditions?

- A. Selection
- B. Iteration
- C. Sequence
- D. Assignment

Q13 (2 marks)

Consider the code:

```
score = 75
if score >= 70:
    grade = "A"
else:
    grade = "B"
```

What value does `grade` hold after execution?

- ○ **A.** "B"
- ○ **B.** 75
- ○ **C.** "A"
- ○ **D.** 70

Q14 (2 marks)

What type of loop is a `for` loop that repeats a fixed number of times?

- ○ **A.** Condition-controlled loop
- ○ **B.** Infinite loop
- ○ **C.** Count-controlled loop
- ○ **D.** Nested loop

Q15 (2 marks)

When is a `while` loop more appropriate than a `for` loop?

- ○ **A.** When you know exactly how many times to repeat

- ○ **B.** When you want to iterate through a list
- ○ **C.** When repetition depends on a condition that may change
- ○ **D.** When you need the loop to run at least once

Q16 (2 marks)

What is the key difference between a function and a procedure?

- ○ **A.** Functions can take parameters but procedures cannot
- ○ **B.** Procedures are faster to execute than functions
- ○ **C.** Procedures can only be used once in a program
- ○ **D.** Functions return a value but procedures do not

Q17 (2 marks)

What is a parameter in relation to a subprogram?

- ○ **A.** The value returned by a function
- ○ **B.** A variable defined inside the subprogram
- ○ **C.** A value passed into a subprogram when it is called
- ○ **D.** The name given to the subprogram

Q18 (2 marks)

What is a local variable?

- ○ **A.** A variable that can be accessed from anywhere in the program
- ○ **B.** A variable declared inside a subprogram, accessible only within it
- ○ **C.** A variable that stores the computer's location
- ○ **D.** A variable that cannot be changed after creation

Q19 (2 marks)

Why should programmers generally avoid using global variables?

- ○ **A.** They use more memory than local variables
- ○ **B.** They make programs run more slowly
- ○ **C.** They can be changed from anywhere, making bugs harder to track
- ○ **D.** They cannot store large amounts of data

Q20 (2 marks)

What is the result of `True and False`?

- ○ **A.** True
- ○ **B.** False
- ○ **C.** Error

- D. None

Q21 (2 marks)

What is the result of `not (5 > 3)`?

- A. False
- B. True
- C. 5
- D. 3

Q22 (2 marks)

What does iteration over a data structure mean?

- A. Deleting all elements from the structure
- B. Processing each element in the structure one at a time
- C. Sorting the elements into order
- D. Copying the structure to a new location

Q23 (2 marks)

Which of the following would cause a runtime error?

- A. Forgetting a colon at the end of an if statement

- ○ **B.** Using the wrong variable name in a calculation
- ○ **C.** Attempting to divide a number by zero
- ○ **D.** Adding two numbers instead of subtracting them

Q24 (3 marks)

Consider the code:

```
x = 15
y = 8
if x > 10:
    if y > 10:
        result = "Both large"
    else:
        result = "Only x large"
else:
    result = "Neither large"
```

What value does `result` hold?

- ○ **A.** "Both large"
- ○ **B.** "Only x large"
- ○ **C.** "Neither large"
- ○ **D.** No value assigned

Q25 (3 marks)

Consider the code:

```
def calculate(a, b):
    total = a + b
    return total * 2
```

```
result = calculate(3, 4)
```

What value does `result` hold?

- A. 7
- B. 14
- C. 12
- D. 24

Q26 (3 marks)

Consider the code:

```
total = 0
for i in range(1, 6):
    if i % 2 == 0:
        total = total + i
```

What is the final value of `total`?

- A. 15
- B. 6
- C. 9
- D. 10

Q27 (3 marks)

What is the result of the expression `3 + 4 * 2 - 1`?

- A. 13
- B. 14
- C. 6
- D. 10

Q28 (3 marks)

A programmer writes:

```
def average(numbers):
    total = 0
    for n in numbers:
        total = total + n
    return total / len(numbers)

result = average([])
```

What type of error occurs when this code runs with an empty list?

- A. Syntax error
- B. Logic error
- C. Runtime error
- D. No error occurs

Q29 (3 marks)

Consider the code:

```
count = 10

def update():
```

```
    count = 5
    return count

result = update()
print(count)
```

What value is printed?

- A. 5
- B. 10
- C. 15
- D. Error

Q30 (3 marks)

Which of the following makes code most difficult to maintain?

- A. Using meaningful variable names
- B. Adding comments to explain complex logic
- C. Using magic numbers instead of named constants
- D. Breaking code into well-named functions

Total: **55** marks • Time: **30 min**

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